

Team Control Number

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Problem Chosen

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Summary Sheet

Summary

Efficient evacuation sweeps are essential for ensuring that all occupants can safely exit a building during emergencies such as fires or gas leaks. Despite the existence of standard operating procedures, responders must frequently navigate complex building layouts with incomplete information and rapidly evolving hazards. Our team developed a mathematical framework that optimizes sweep operations in multi-floor buildings while balancing responder safety, sweep completeness, and overall time efficiency applied to all kinds of complex building layouts.

Our modeling process began from the foundational behaviors of individual responders. We identified critical sweep parameters, including rules of movement, sweep verification requirements, responder count, occupant types and hazard conditions, that fundamentally shape sweep performance. Using these criteria, we constructed an **agent-based simulation** implemented on a **discretized grid system**.

In order to analyze baseline performance, we first applied our model to the provided one-story office floor plan featuring three rooms on each side of a central hallway and two exits. By simulating all feasible movement paths under ideal conditions, we determined the **minimum clearance time** for two responders and identified the optimal division of labor and sweep direction.

We then extended our model to more complex scenarios, designing two additional multi-floor building layouts with different responder counts and occupant types like office workers and childcare facilities. For each case, our model outputs (1) the **optimal sequence** of rooms to check, (2) ideal **responder starting locations**, and (3) the **minimum achievable sweep time**. Our analysis demonstrates that layouts with longer hallways or non-linear floor structures benefit significantly from parallelized responder deployment and redundancy assignment based on room risk level.

To incorporate real-world constraints, we refined our model to include smoke-induced visibility decay, dynamic occupant movement, and hazardous zone propagation.

Finally, we developed a practical evacuation-sweep strategy informed by our modeling results. This includes guidelines for responder assignment, recommended technologies for varying building types, and redundancy protocols that ensure sweep completeness without sacrificing efficiency.

Keywords: Agent-Based Modeling, Evacuation Sweep Optimization, Redundancy Planning, Discretized Floor Simulation, Building Layout Analysis

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1. Introduction

1.1. Background

Nowadays, as architectures displays an increasing trend in height, they vary in height, size, and shape. Concerns have been raised about natural and anthropogenic hazards like fire, gas leaks, and earthquakes threatening the lives of the people trapped in these buildings who fail to escape in time. In light of this, an efficient, organized, and readily available evacuation strategy is necessary for each of these sites. This is why we have responders, the people who are trained professionally to respond to these kinds of incidents, to ensure, by sweeping all corners, that no occupants are left inside the perilous building and the number of fatalities and injuries is minimized.

1.2. Problem Restatement

Although the responders have been trained systematically, it still remains doubtful if their current method of addressing this kind of emergency situations is of maximum efficiency. Given that occupants' safety and responder efficiency are prior to the rest, a set of optimized strategies should be developed. We should devise a viable method that is always applicable even in the face of high pressures or limited information available. We need to figure out, for different types of hazards, how many responders are needed, what jobs should be allocated to them, how can they sweep the whole floor most efficiently, etc. Then, an emergency should be simulated in order to help refine the set of strategies.

1.3. Assumptions and Justifications

Assumption: All regular doors are identical (in terms of fire and smoke spread and gas leak), except fire-proof doors which fire cannot spread through, and is harder for smoke and gas to leak through

Justification: Building codes require standardized door performance within a fire-resistant rating category, allowing consistent modeling of barrier effects.

Assumption: Firefighters conducting searches are assumed not to alter the rate of fire spread or gas dispersion. Actual firefighting operations are treated as external and already factored into a stabilized hazard-spread rate prior to the sweep.

Justification: The problem focuses on optimizing sweep strategies rather than suppression techniques. Decoupling these processes isolates the variables relevant to responder movement and occupant safety. In addition, fire evacuation models like FDS-Evac assume hazard parameters are independent of search activity for computational simplicity. [1]

Assumption: The building is divided into uniform square tiles of side length l , each representing a localized, homogeneous environment for hazard spreading and occupant movement.

Justification The model encourages modeling movement and hazard dynamics discretely, which aligns with common grid-based approaches in fire and gas simulation literature.

Assumption: In a room without natural ventilation (no open windows), oxygen is the limiting factor of fire spread instead of amount of flammables in the room.

Justification: [2]

Assumption: Communication in sweep operations in the study primarily rely on non-verbal, self-contained search patterns (detail seen in 3.3.3) that do not require continuous radio

communication.

Justification: Responders coordinate primarily through predefined sweep patterns instead of dynamic verbal instruction. Communication between firefighters and occupants relies on physical senses in auditory range (within 2-3 tiles) and visual range (computed using Koschimieder's Law).

Assumption: The volume that smoke particles displace in the air is almost negligible.

Justification: During building fires, dangerous smoke conditions typically involve PM2.5 (particulate matter under 2.5 micrometers) and toxic gases like CO and HCN. In typical building fires, $1000 \text{ mg} \cdot \text{m}^{-3}$ is considered extremely high concentrations of PM2.5, and only represents only 0.0001% of the air volume [3], and CO and HCN cause incapacitation at parts-per-million concentrations [4]. Thus the smoke spread model is very similar to the fire model, but different from the gas leak one.

Assumption: Fire mostly damages people by direct contact or producing harmful smoke, fire not very close to the person have negligible direct damage (caused by heat, not counting smoke) compared to these 2 sources.

Justification: An estimated 50-80% of all fire deaths result from smoke inhalation injuries rather than burns [5]. Direct damage from fire mostly depends on radiant heat transfer, which is inversely proportional to the square of the distance. This paper also uses a tile-based modeling, so evacuates can stay away from fire not on the same tile, making the only primary damage sources direct damage from the fire on the same tile and smoke produced by all nearby fires.

Assumption: Fire does not spread through empty floor.

Justification: Modern buildings usually use ceramic or concrete floors which is fire-resistant and non-combustible, the main way fire spreads is through flammable objects in close range, such as wooden furnitures.

Assumption: In a single-source gas leak, there is only one major leak path to any room graph-wise (if 2 paths pass through the same rooms in the same order, it is considered the same path)

Justification: When the rooms of a building layout is drawn as the vertices of the graph and the rooms' direct connectivity as edges, most of the buildings' graph has a tree structure with no "loops" in the graph, meaning that between any 2 rooms there is only 1 unique path room-wise. Although it is common for buildings like shopping malls to have a "hole" in the middle of the building for aesthetics and placing escalators, the entire corridor containing the "hole" is considered as one room (since it is not separated by doors) and the hole does not show up in the graph.

2. Variables and Parameters

Note that all parameters and variables used in the text are listed in each subsection in alphabetical order, and is totally irrelevant to the order they appear in the text.

2.1. Generic (For All Emergency Types)

- l : side length of a "tile" in the model (unit: m)
- $H(a)$: the relative and inherent health value for an individual, 0 indicates one has deceased, 100 is set to be the highest value one can reach.
- HP : the relative value tracking a specific individual's health status.
- $v_{\text{firefighter}}$: the speed of the firefighter

2.2. Fire Model

2.2.1. Fire spread

- $\omega(r, t)$: the ratio of the volume of remaining oxygen in room r at time t to the volume of oxygen the room initially holds
- $P_f(\omega)$: rate of fire spread to an adjacent tile per second (unit: time^{-1})
- $P_{f\text{-door}}(\omega)$: rate of fire spread to an adjacent tile with a door between per second (unit: s^{-1})
- $P_{f\text{-self}}(\omega)$: rate of fire growth on its tile per second if flammables on the tile is sufficient (unit: s^{-1}), should be larger than 1 when ω is sufficiently high
- r_0 : the base unit for fire size based on rate of smoke production, see model mechanisms
- R_c : rate at which fire of size r_0 consumes oxygen
- $r(c, t)$: the amount of fire burning on tile c at time t , unit is r_0
- $r_{\max}(c)$: the theoretical maximum amount of fire, which is when every inflammable that is in contact of oxygen is burning, on tile c , unit is r_0 . On empty floor tiles it is equal to 0 or a negligibly small value because non-wooden floor tiles are fireproof and cannot burn.

2.2.2. Smoke spread

- P_s : rate of smoke spread to an adjacent tile per second (unit: time^{-1})
- $P_{s\text{-door}}$: rate of smoke spread through the door. Should be lower than P_s but faster than the rate at which fire passes through doors [6]
- $P_{s\text{-self}}(c)$: rate of smoke spread to itself when it borders 4 tiles with no separation. Should be less than 1 as smoke naturally disperses
- s_0 : unit amount of smoke. Produced by a fire of size r_0 each second
- $s(c, t)$: concentration of smoke caused by fire on tile c at time t

2.2.3. Visibility

- β_{ext} : coefficient that quantifies the total light loss due to smoke (scattering and absorption), unit is m^{-1}
- d : distance of visibility, unit is m
- $-\ln(\varepsilon)$: minimum contrast the human eye needs to distinguish an object from its background (typically 0.05 or 5%).
- K_m : mass-specific extinction coefficient, unit is m^2/kg

2.3. Gas Leak Model

2.3.1. Units

- γ_1 : ratio of methane gas in the room at saturation
- V_0 : volume of room with 1 tile area
- v_0 : base gas leak rate that changes γ_r from 0 to γ_1 in a room of volume V_0 in unit time

2.3.2. Gas spread

- c : concentration of gas, unit is mass/volume
- λ : ventilation rate, unit is inverse time
- G : flow rate, unit is mass/time
- V : room volume

3. Model Creation

3.1. Model Objectives

The primary objective of our model is to minimize the total time required to complete a full building sweep while ensuring the safety of all occupants and responders. A secondary objective is to optimize outcomes of the rescue operation after hazard dynamics, such as fire spread, smoke accumulation, and toxic gas concentration is taken into consideration. This requires an optimum sweeping strategy that gives the most ideal combination of operation outcomes including occupant survival probability, number of casualties, survivors' health status, etc. Note that minimizing operation time is not part of the secondary objectives, for time consideration simply contributes to less spread of hazards and thus higher survival possibility of occupants.

3.2. Model Mechanism

3.2.1. Fire model mechanism

3.2.1.1. Fire spread

When considering the mechanisms during fire events, we decided to divide the whole map into small square tiles that has length l customized for each scenario based on the corresponding architectural composition.

The fire can gradually self-extinguish or spread to adjacent tiles. The rate for the fire to maintain its combustion and spread to neighboring tiles are given by $P_{f\text{-self}}(\omega)$ and $P_{f\text{-door}}(\omega)$, respectively. Note that the rate here is the macroscopic net outcome resulting from the constantly accumulating microscopic probabilities. Also, oxygen depletion is taken into account: a combustion constantly consumes oxygen to support combustion, and without an oxygen provider, assumingly a window or other ventilation, the fire will eventually die down. When a occupant is in a tile on fire, he or she loses HP value constantly at a rate dependent on ω , which can indicate how much oxygen the combustion has left to consume, through which the severity the fire can be deduced.

The size of the fire $r(c, t)$ can be described by the fire spread from all adjacent tiles, either separate by nothing or by a door, and the tendency for the fire to stay on its own tile. If the sum of all contributing factors is larger than $r_{\max(c)}$, then the size of the fire will be $r_{\max(c)}$ since the amount of combustibles with contact to oxygen is a limiting factor. Thus $r(c, t) = \min\left(r_{\max(c)}, \sum_{c' \text{ is directly adjacent to } c} (P_f(\omega(c')) \cdot r(c', t-1)) + \sum_{c' \text{ is adjacent to } c \text{ by a door}} (P_{f\text{-door}}(\omega(c')) \cdot r(c', t-1)) + P_{f\text{-self}} \cdot r(c, t-1)\right)$

The rate that fire spreads to adjacent tiles, through doors, and grows on its own tile are both proportional to ω . Let P_{f0} , $P_{f\text{-self}0}$, and $P_{f\text{-door}0}$ represent the ability for fire to spread to adjacent tiles, grow on its own tile, and spread through doors when the room is at full oxygen ($\omega = 1$), then $P_f(\omega) = P_{f0} \cdot \omega$, $P_{f\text{-self}}(\omega) = P_{f\text{-self}0} \cdot \omega$, and $P_{f\text{-door}}(\omega) = P_{f\text{-door}0} \cdot \omega$. Fire also consumes oxygen at the rate proportional to its size. So $\omega(r, t) = \omega(r, t-1) - \sum_{c \in r} R_c \cdot r(c, t-1)$.

3.2.1.2. Smoke spread

Regarding the modelling of the spreading of smoke, a deadly byproduct of fire, we used a similar fundamental method: to divide the site into small homogeneous tiles and consider the smoke concentration in each one. Differing from fire, the smoke will not self-extinguish, it only becomes dilute enough through time or ventilation to be harmless. Although it does spread to

neighboring tiles, its total volume remains constant when we exclude those constantly produced by the existing fire and focus on the existing part. Therefore, we consider the two parts of smoke separately. For the existing smoke, we consider its rate of diffusion into neighboring tiles while keeping total mass constant, which may be affected by how the two tiles are separated, such as by a door, a wall, or nothing. Note that change in air pressure is not taken into consideration, for the deadly level of toxic smoke is negligibly small to make any difference in air pressure.[7] For smoke newly produced, we assume all its concentration remains in the original tile in this current unit time, and consider it as existing smoke in the next.

Since the mass and volume of existing smoke is conserved, $P_{s\text{-self}}(c) = 1 - \sum_{c' \text{ is directly adjacent to } c} (P_s) - \sum_{c' \text{ is separated by a door to } c} (P_{s\text{-door}})$. Meaning that $P_{s\text{-self}}$ is larger the more walls or doors border the tile.

Similar to the fire spread model, the smoke concentration can be calculated by $s(c, t) = P_{s\text{-self}}(c) \cdot s(c, t-1) + \sum_{c' \text{ is directly adjacent to } c} (P_s \cdot s(c', t-1)) + \sum_{c' \text{ is separated by a door to } c} (P_{s\text{-door}} \cdot s(c', t-1))$. Note that for smoke concentration, the upper limit can be ignored as even very concentrated smoke has a negligible actual volumetric concentration, making the physical limit basically unreachable.

When applying to building layouts with multiple floors, the regular stairs (not outdoor fire-proof stairs) can be considered as a connecting tile between 2 tiles on different floors, allowing fire and smoke to spread through floors. In simulation, this can be done by updating the tile adjacency table and doing the calculations exactly like before.

3.2.2. Gas leak model mechanism

3.2.2.1. In a single room

In a room with volume V (proportional to the number of tiles in the room), and gas leak rate into the room G , the rate of change in concentration can be calculated by

$$\frac{dc}{dt} = \frac{G}{V} - \lambda c$$

For multiple leakage sources in the same room, the rate of change in gas concentration is the sum of rate of change of concentration of each individual leak, or

$$\frac{dc}{dt} = \frac{\sum G}{V} - \lambda c$$

3.2.2.2. Across multiple rooms

Assuming only one source of gas leak and only one valid path (room-wise, multiple paths through the same rooms in the same order counts as one path) from the source room to each room, then for each room r that is not the source room, $\sum G(r) = c(r') \cdot \sum G(r')$, where r' is the room which the gas is in before leaking into room r .

3.2.2.3. Discretization for simulation

During each “tick” of the time unit, $\sum G$ of each room is first updated recursively by

$$\sum G(r, t) = c(r', t - 1) \cdot \sum G(r', t - 1)$$

and the base case $\sum G(r_0, t) = G_0 \forall t$ (where r_0 is the source room of the gas leak). Then the concentration of each room can be updated:

$$c(r, t) = c(r, t - 1) + \frac{\sum G(r, t)}{V(r)} - \lambda(r) \cdot c(r, t - 1)$$

3.3. Sweeping Strategies Candidates

Sweeping strategies in our model operate in parallel with balanced speed, coverage reliability, and responder safety. Although the conceptual structure of primary and secondary searches remains consistent across all hazards, the execution, priorities, and constraints differ substantially between fire and gas leaks due to the distinct physical properties and risks of each emergency type. Our model reflects this by assigning hazard-specific movement rules and prioritization.

3.3.1. For Fire

Primary searches under fire emergencies prioritize the immediate rescue of occupants at risk of incapacitation from flame exposure, smoke spread, or extreme heat. Since visibility is often near zero, responders rely on tactile navigation and rapid entry techniques.

This is represented in our model through:

- high initial responder speed
- strong penalization for tiles with high flame ratio $r(c, t)$
- reduced speed under low visibility using $d = \frac{-\ln(\epsilon)}{\beta_{\text{ext}}}$
- minimal sweep-verification time

Dynamic re-routing occurs when spread probability $P_{\text{fire}()}$ indicates imminent hazard escalation.

Secondary fire searches occur after partial suppression, when visibility improves and temperatures stabilize. This phase focuses on complete verification of each room, especially small or obstructed areas.

3.3.2. For Gas Leak

Gas-leak environments differ fundamentally from the fire. There are no flames or temperature spikes, but gases may be odorless, invisible, rapidly or slowly accumulating, and capable of causing delayed but severe incapacitation. Gas dispersion patterns are governed by diffusion and door-based distances d_r , resulting in a more spatially predictable but physiologically dangerous hazard for occupants.

Primary searches under gas-leak conditions focus on quickly locating occupants who may be unaware of the leak or beginning to experience dizziness due to the leak. Unlike fire, visibility doesn't impair speed, but responder exposure time is limited by cumulative gas toxicity, which the model accounts for through time-dependent health penalties.

Secondary gas-leak searches ensure that no rooms with partial gas infiltration were missed. Responders recheck areas with high leak potential, ventilation junctions, and rooms with slow diffusion rates.

3.3.3. Sweeping Search Types

1. Left-hand & Right-hand Search

Firefighters enter a room and immediately commit to either the left wall or the right wall. Keep one hand or tool in continuous contact with that wall, following it around the room until they return to the entry point.

Maintaining wall contact prevents disorientation and ensures all room boundaries are traced even in complete darkness. The technique is mainly used when searching for unconscious victims or when mapping gas accumulation near enclosed spaces.

2. Oriented Search

Used in moderate-to-high visibility areas, firefighters track landmarks such as windows, doors, or large furniture to maintain spatial awareness and remain oriented. One firefighter acts as the “oriented firefighter”, staying near a clear reference point such as a doorway, window, or major piece of furniture. The other firefighter quickly searches deeper into the room.

3. Anchored Search

In low-visibility conditions, responders keep one hand or foot in contact with outer walls or stable reference points while extending their reach into room centers when safe. This maintains orientation and allows coverage of central floor areas without losing the exit.

4. Vent Enter Isolate Search (VEIS)

Firefighters ventilate through a window, climb in, immediately close the door to isolate the room from the fire, search quickly, then exit through the same opening.

3.4. Basic Sweeping Strategy

We present a sweeping strategy based on the “wall-follower” strategy for maze solving. In a simple building layout where all rooms are connected to a central corridor in the middle like the example given in the problem, following the wall will allow the entire map to be walked through and all rooms to be cleared. After a room is cleared, the room is marked with an easily recognizable object like a sticky note, so that other responders know that the room does not need to be swept again. This is a key advantage of the wall-following strategy, since once a responder sees a room with a sticky note marking it as clear, they know that all rooms after this are clear as well by the same responder who cleared this room, making seeing a cleared room a sign that the specific responder’s sweep is finished.

However, the “wall-follower” will not search certain rooms in more complex layouts. When the corridor is a ring and there are rooms on both sides, or when the corridor is a structure with multiple rings like a figure-8 shape, a wall-following strategy will only search the rooms on the outer side of the building. Thus to make the strategy more adaptable, at least one responder need to wall-follow the outer side, as well as each of the “hole” in the corridor. These are showcased using the more complex layouts we chose to apply the model to.

In the pre-refined model, smoke reducing vision is not considered, and tiles are cleared as long as a responder can spend 2 seconds with the tile in their vision (responders also have 360 degree full vision because they can turn around). For most of the cases, this means that a room can be cleared just by a responder stepping 1 tile into the room and spending 1 second there.

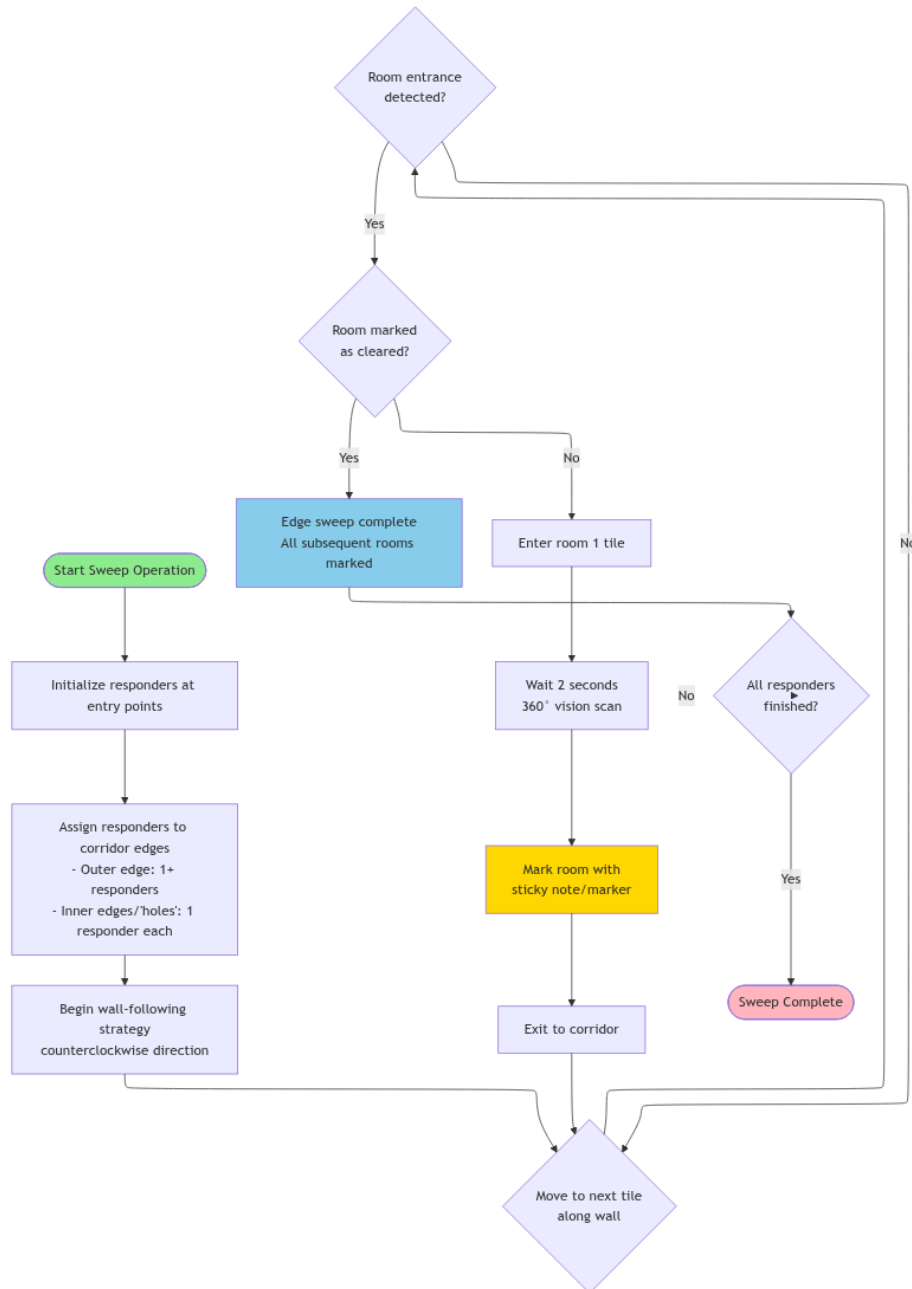


Figure 1: A flow chart for rules of movement for each responder

4. Layout Design

To simulate the model, we used different scenarios of various building layouts to make sure that the model works universally. We can either acquire layouts information of buildings from the Internet or design them according to buildings in reality.

4.1. Simple Layout Clarification

We used the model with 6 rooms and a hall given in the problem sheet for a simple layout. Slight alternations were made to make it adapt to the 2 meter by 2 meter grid lines and fit our model more.

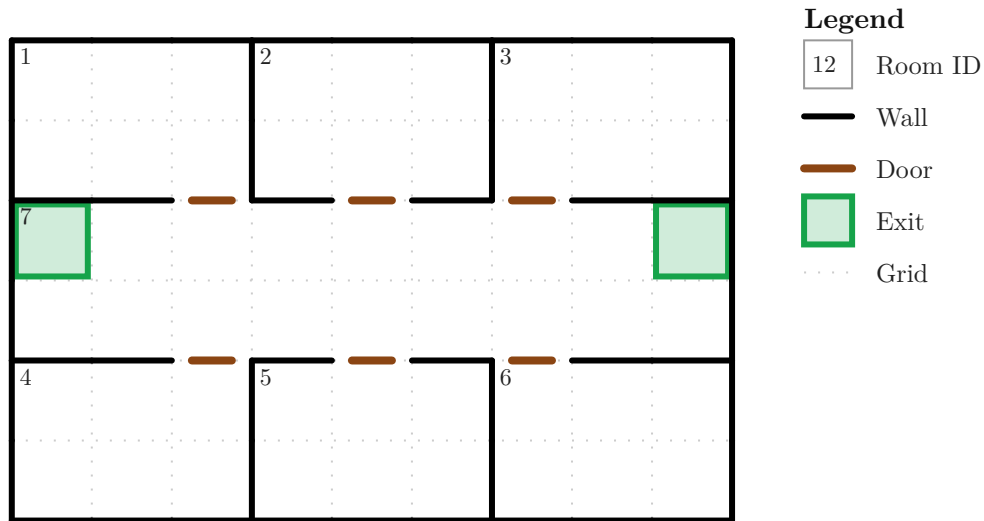


Figure 2: The simple layout given in the problem, adapted for our model, with grid lines

4.2. Complex Layout Design

For more complex situations, we referred to some buildings around our community, a school and a mall's designs were picked as the origins for the layouts.

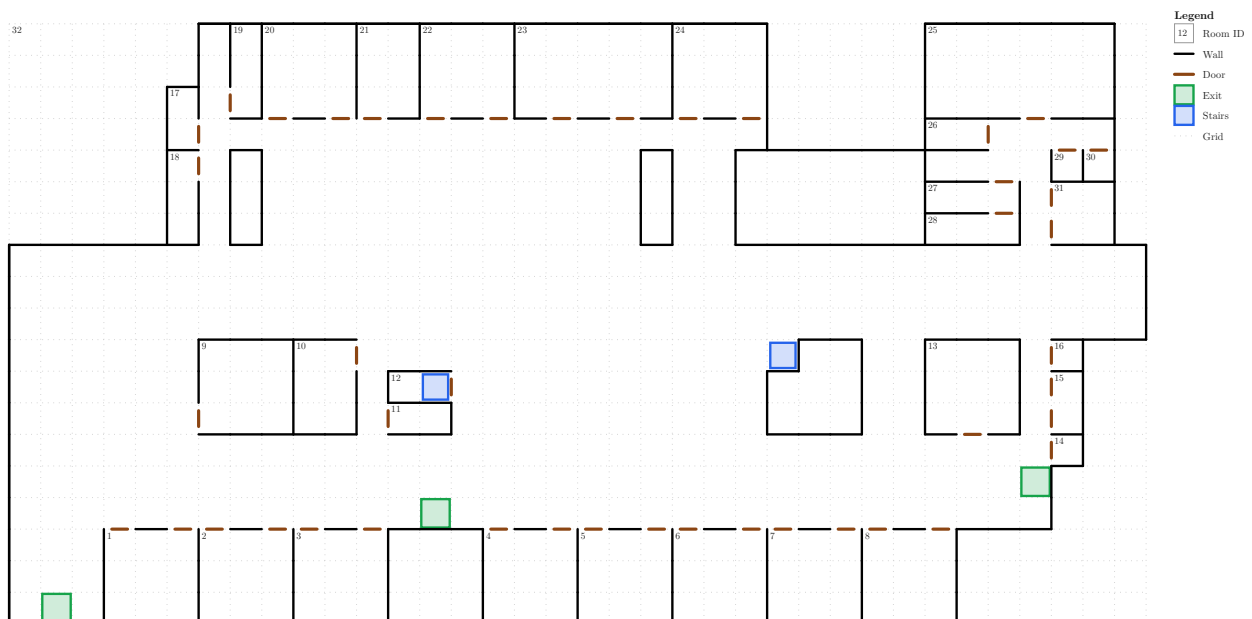




Figure 5: This school layout has a large central corridor which has a “hole” in the middle for the second and third floors, making the wall-following strategy require adaptation. Floor 1, 2, and 3 from top to bottom.

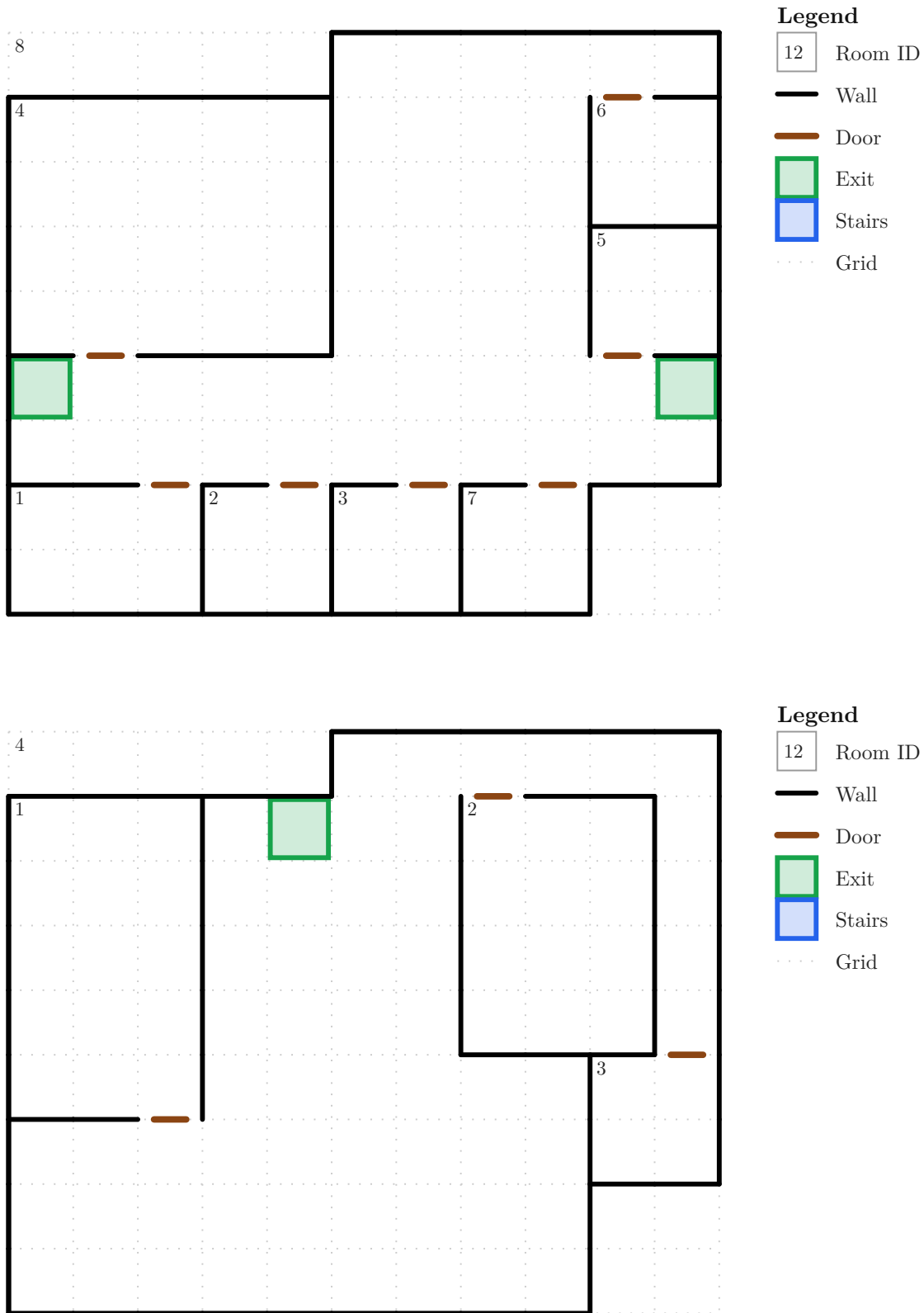


Figure 7: A mall layout. The floor plan on top is an office floor, and the floor plan on the bottom is a restaurant floor.

5. Model Application

5.1. Chosen Constants

To ensure our evacuation simulation reflects realistic fire with triggering smoke and gas dynamics, we calibrated all hazard-related parameters using values consistent with building-scale fire protection engineering and simplified compartment fire models. The constants were selected to maintain physical plausibility within the discrete-step simulation environment, and appropriate relative scaling between different hazards. Below we describe and justify the final constants adopted for our model.

5.1.1. Generic Parameters

- $l = 2\text{m}$ (width of tiles): 2 meters is used in this model because it is accurate enough, and not too small that the diamond-shaped spreading caused by square grids become non-realistic.
- $v_{\text{firefighter}} = 1\text{m/s}$: firefighters move slightly faster consistent with trained responder movement in lightly smoky corridors, this also works well with the 2 meter tile width, meaning that firefighters move 1 tile each 2 seconds

5.1.2. Fire Model Constants

- $P_{f,0} = 0.08$: baseline probability of fire spread to an adjacent tile. Typical compartment fires double in size in minutes, so per-second spread of 0.08-0.12 is realistic for small rooms.
- $P_{f\text{-self},0} = 1.25$: fire grows faster on its own tile than it spreads, thus a rate > 1 ensures the sustainable growth while oxygen is available.
- $P_{f\text{-door},0} = 0.015$: doors restrict flame spread significantly. Door transmission is about 20% of wall adjacency, aligning with common fire-door leakage behavior. [8]
- $r_0 = 1.0$: normalizes fire intensity to one “unit” per tile
- $R_c = 0.05$: a full intensity fire reduces available oxygen by about 5% per second in a closed 1 tile room. This makes oxygen the limiting reactant after 20 seconds, realistically slowing the fire.

5.1.3. Smoke Model Constants

- $S_0 = 0.20$: each unit of fire generates moderate smoke per second
- $P_S = 0.18$: smoke spreads faster than flames. A 0.18 per second rate allows the corridor smoke buildup in less than 30 seconds
- $P_{s\text{-door}} = 0.10$: doors restrict but do not prevent smoke leakage. 10% per second is realistic for under-door gap airflow based on pressure-driven smoke migration, confirming the [6] criteria.

5.1.4. Gas Leak Model Constants

- $\lambda = 0.04$: gas spreads more slowly than smoke due to density and lower thermal buoyancy. A 0.04 per second rate allows gradual filling of a room in about 25 seconds if unventilated.
- $\lambda_{\text{door}} = 0.12$: gas flows 3 times faster through door gaps due to pressure equalization

5.2. Simple Layout

Due to the layout’s relative simplicity, we choose to place fewer amount of responders since using too much can actually cause chaos between themselves and is also not favored economically.

Using 2 responders starting at exits (1, 4) and (9, 4) (all coordinates are 1-indexed), the

responder on the left clears rooms 1, 2, then 3 in this order, spending 28 seconds. The responder on the right spends 30 seconds clearing rooms 6, 5, and 4 (in this order). At 34 seconds, both responders saw that the next room they will sweep has already been cleared, clearing the entire building in 34 seconds.

5.3. Complex Layout

We then applied the models on buildings of a school and a mall. As aforementioned, we want the sweep to be effective, so we do expect that as map complexity increases, we need more trained responders to cope with the situation. From figures depicted above, we see that the school's map is more complex and thus more responders are placed in the school layout to combat the situations.

5.3.1. School simulation

For floors 2 and 3, the outer edge has a length of 107 tiles, taking 214 seconds to do a complete walk. It passes through 36 rooms, each adding 6 seconds to the clearing (enter, clear, and exit the room, with 2 seconds for each action) with a total of 430 seconds. The inner edge is 62 tiles long, and has 5 rooms, with a total of 154 seconds. Using 3 responders for each floor, with 1 clearing the inner edge and 2 clearing the outer edge, the optimal total time should be $\frac{430}{2} = 215$ seconds since each responder on the outer edge clears half of the room. This optimal time can be achieved by having one responder enter in one of the stairs marked in blue, and the other entering from the evacuation stair marked in green.

For floor 1, the outer edge has a length of 99 tiles with 26 rooms, with a total of 354 seconds. There are 3 inside edges or "islands" in the main corridor: rooms 9-10, 11-12, and 13, taking 48, 36, and 38 seconds to clear, adding up to 122 seconds not counting travel time between them. If 2 responders are deployed for clearing the outer edge, which will take $\frac{354}{2} = 177$ seconds, one responder clearing the entire inside will finish in about the same time.

5.3.2. Mall situation

The office floor's corridor's outer edge is 31 tiles long, with 6 rooms, giving a total of 98 seconds. The 2 entrances are also placed efficiently so that a responder starting from each exit can finishing the sweep in 52 seconds, slightly over half of the total time.

For the restaurant floor, 2 responders take 30 seconds to clear the floor, which is much faster since this floor only has 3 rooms.

6. Model Refinement

6.1. Responder Movement

Responders can move to a diagonally adjacent tile as well as an orthogonally adjacent one. This is a more realistic simulation of human walking, making walking through diagonal lines more efficient on distance and time. Walking one diagonal tile should take $\sqrt{2}$ seconds (when the walking speed is 1 tile per second), to implement this in simulation, each responder has a `distance-remaining` attribute that increases by 0.5 per second, moving orthogonally deduces 1 from it, and moving diagonally deduces $\sqrt{2}$. After every move, the value should remain non-negative.

6.2. Reduced Visibility from Smoke

Visibility in smoke-filled environments caused by fire is a decisive factor for both occupant survival and the effectiveness of responder sweeping strategies. As smoke accumulates, suspended particles scatter and absorb the light, reducing the contrast needed for human eyes to recognize objects and navigate safely. Our model follows the standard formulation of Koschmieder's Law, which relates visibility distance to the optical properties of smoke.

Koschmieder's Law [9], in its standard form, states that the maximum distance at which an object can be distinguished from homogeneous (uniform) atmosphere is

$$d = \frac{-\ln(\varepsilon)}{\beta_{\text{ext}}}$$

where:

- ε : minimum contrast threshold detectable by the human eye (commonly $\varepsilon = 0.05$, representing 5% of contrast)
- β_{ext} : extinction coefficient, which quantifies the total attenuation of light due to scattering and absorption by smoke particles (treated as a single, constant value)

Whereas in a smoke simulation that uses tiles, the smoke concentration (and therefore the extinction coefficient) changes at every point r along the line of sight. The total reduction in contrast is the cumulative effect of attenuation over the entire distance.

This process is described by the **Beer-Lambert-Bounguer Law**, which, in its differential form, shows that the change in light intensity dI over an infinitesimal path length dr is proportional to the local extinction coefficient $\beta_{\text{ext}}(r)$ and the current light intensity I :

$$\frac{dI}{dr} = -I \cdot \beta_{\text{ext}}(r)$$

To find the maximum visibility distance d , we must integrate the extinction coefficient along the path. The distance d is the point where the total accumulated optical depth (the integral of the extinction coefficient) reduces the initial contrast below the minimum detectable threshold ε .

The relationship for maximum visibility d in a non-uniform medium is given by:

$$\int_0^d \beta_{\text{ext}}(r) dr = -\ln(\varepsilon)$$

In our model, a tile-based simulation, this integral is typically approximated as a sum:

$$\sum_{i=1}^N \beta_{\text{ext},i} \cdot \Delta r_i = -\ln(\varepsilon)$$

Furthermore, β_{ext} is derived from the smoke concentration on a tile:

$$\beta_{\text{ext}(c,t)} = K_m s(c,t)$$

K_m is the smoke specific attenuation constant, representing how much visibility is reduced for every kilogram of smoke particles suspended in a cubic meter of air. A commonly referenced value is around $8700 \text{ m}^2/\text{kg}$ for smoke generated by flaming fires. [10]

The linear approximation $\beta \propto s$ is widely used in fire-safety engineering since particulate density is the dominant driver of visibility reduction for the concentration range typical of fires emerging in buildings.

6.3. High-Risk Area Prioritization

If the building is evacuated because of fire, the room where the fire started in is first visited and cleared by 2 responders (multiple responders might be needed to quickly clear this room because it likely has thick smoke that reduces visibility, requiring each tile to be cleared independently). Then the responders split up and starts sweeping their “edge” until the entire edge has been swept. The other edges of the corridors are dealt with like before. In this way the rooms near fire, which is in relatively more danger, is also being cleared earlier. During a gas leak, the same procedure can be done, to evacuate people closer to the leak earlier and make the people in most danger more aware of the hazard.

6.3.1. Further optimization for multiple floors

The floor where the hazard started, as well as the floor above and below it, is first being quickly swept using a larger number of responders, then a standard number of responders are left for secondary sweeps in these floors for guiding evacuating people from floors above, while the rest goes to where sweeps are slower and more responders are needed.

6.4. Time-Varying Hazards

The spread of fire is now properly taken into consideration. Responders cannot directly walk into tiles where the fire is sufficiently large, and spreading fire creates more smoke that spread to more parts of the map, reducing overall visibility. The spread of fire and smoke can be simulated in the way mentioned in Section 3.2.1.

6.5. People Moving During Evacuation and Redundancy

It is possible that some people evacuating will not be seen by moving responders, and they might end up in hazard zones due to lack of guidance. Therefore the hazard zones where evacuation routes pass through should be a high priority region for secondary sweeps, and stationary responders should be deployed at these places if possible.

6.6. Use of Technology

The effectiveness of emergency evacuation sweeps in multi-floor buildings relies heavily on timely information, responder coordination, and accurate verification of cleared spaces. While traditional sweep models depend on trained personnel following predetermined paths and protocols, the integration of modern technologies such as sensors and building automation can significantly enhance both the speed and safety of these operations. In our model, we examined building layouts including single-story office buildings with multiple exists, multi-floor layouts and rooms with varying occupant profiles. It becomes clear that technology can serve as a force multiplier, improving awareness and reducing human error.

Public areas are usually required to have fire prevention equipments set within. For instance, there are fire alarms and other electronic devices that detect and remind everyone of the situation in our scenario of the school layout, which could potentially enable most of the people to escape from the hazard by themselves. Furthermore, firewalls are also set there and could be dropped down when there is a fire emergency, holding back the spread of smokes and fire to some extent. In terms of our model, occupants relatively far from tiles on fire gain extra time to react and escape when there are fire alarms available, and fire and smoke on tiles surrounded by firewalls will have a significant decrease in rate of spreading to neighboring tiles.

6.7. Application of Refined Models on Complex Layout

This refined model is tested on the school layout, where fire starts in room 24 of the third floor. Like in the simpler model, 9 responders are deployed initially. However 4 are deployed on the third floor, 4 on the second floor, and 1 on the first floor. The simulation shows that the third floor was cleared in 202 seconds, compared to the 215 seconds in the simpler model. Despite smoke blocking vision in certain rooms, prioritizing the fire origin floor by deploying 5 responders instead of 3 sped up the sweep significantly and outweighed the effect of reduced vision. The second floor was cleared in 174 seconds, which is much faster than the sweep time of the third floor because smoke only spread to the second floor until when most of the sweep is finished. Although 1 responder each from the second and third floors goes to the first floor after clearing their floors, the sweep time of the first floor is still significantly longer at 264 seconds compared to the 177 seconds using the model before refinement.

Overall, the sweeping time used by the refined model is actually longer than the pre-refine model, with the first floor being the bottleneck taking 264 seconds. However, the second and third floors, especially the rooms near where the fire started, was cleared very early and with extra redundancy, meaning that the people who need the most urgent evacuation gets rescued much faster than the pre-refine model, increasing the evacuation's survival rate. When the first floor was cleared, the smoke has not yet reached dangerous levels in large areas, therefore the evacuation using the refined model is overall more safe.

7. Evaluations

7.1. Overall Strengths

We successfully devised a set of adoptable strategies that confronts break-outs quickly and decently. We made sure that responders are able to sweep all corners in limited time in most buildings and bring occupants to safety. Reduced time also that there will be a decline in total health loss, which means that there will be less injuries and hopefully no fatalities. We also used the models we created to gain an intuitive perspective on how many responders are needed for a building. This is due to the fact that we can roughly estimate a building's capacity and its general structural complexity only from its appearance.

7.2. Limitations and Improvements

7.2.1. Isolated Islands

We adopted the “wall-follower rule” as the basic sweeping strategy, however, there is a constraint when applying the rule. To facilitate understanding, consider two concentric circles as walls, if one always sticks to the wall on the outskirts, the things around the smaller circle would be ignored. In this case, the small circle is the isolated island that is unreachable.

There are several methods to address this. Most effectively, if a map is given, it would be clear which part would be isolated and should be paid attention to. Another scenario is that no maps was provided and limited information was given to the responders. In this case, we need one responder to act as a commander. While other responders head into the place of hazard with locating and communicating equipment, the commander receives information displaying what routes responders are taking and what placed have they been to. Thu, when the commander see that all routes taken seems to avoid a certain region on the map, there is probably an island that responders missed. The commander system is not just for the no-map case, it can also be used in other scenarios since this ensures that every region has been swept.

Another method that does the job well is to have a few members in the responders' team, who cooperates with other responders that follows the orthodox strategy, to track the inner parts of the building. In some cases, this might be a slightly weaker response compared to the last method introduced in terms of efficiency since the people sweeping the inner parts are not machines and don't do as perfect. Though the whole sweeping is still guaranteed to occur more evenly throughout the building. In some cases where gas leakage occurs, this is a better alternative than the previous one, since nobody knows exactly what kinds of gases are leaked. It is perilous to rashly equip responders with electronic devices that might cause explosion when releasing and receiving signals. Not to mention that the equipment may fail to deliver signals in thick gases.

7.2.2. Qualitatively versus Quantitatively

The models are all modelled qualitatively, which means that though they can indeed give useful results that organizations can refer to, the precision and risk perception ability had been compromised. Thus, we also devised a theoretically possible method to quantify most things. That being said, we use health points to quantify the health, which has an initial value that precedes 100 and is decreasing during the hazard. It might also be more quantitative to derive speed functions and damage functions. We can also randomly assign occupants in rooms or halls so that the small place they are in resembles an age and health structure just like the society.

An example is done for the estimation of the health equation. Starting by figuring out the shape of the desired health function. We found that the Gompertz-Makeham law of mortality describes the correlation of dying chances and age for adults [11] and it manifests the chance of dying after 20 is increasing exponentially. Thus we can model the health to peak at age 20 and health 100 (as defined previously). We expect a concave down curve for health at first since after 20 death chances are exponentially increasing. At around 70, health deteriorates most significantly so 70 should be an inflection point. At the age of 100. So for $a > 20$, we adapt the Gompertz-Makeham in some way that these expectations can be achieved. While at an age before 20 is another increasing function that has an inflection point at age 10 (where fastest growth occur) [12].

According to these data, a possible function of health is derived:

$$H(a) = \begin{cases} H_1(a) = -\frac{1}{50}a^3 + \frac{3}{5}a^2 + 20 & (0 \leq a < 20) \\ H_2(a) = 100\left(\frac{\mu(20)}{\mu(a)}\right)^{\frac{1}{3}} & (20 \leq a \leq 100) \end{cases}$$

where $\mu(a)$ is in the Gompertz-Makeham form:

$$\mu(a) = 5 \times 10^{-4}(1 + 0.1e^{0.1a}), (20 \leq a \leq 100)$$

The speed and damage function can also be found analogously as functions of preceding functions.

This method is a way to check if a sweeping method is feasible by quantifying the amount of health loss. However, it is often more time-consuming to research everything and requires calculating everything precisely. This would be a choice when very specific data is needed.

8. Conclusion

In a nutshell, after investigating nuances in different models, we came up with a set of strategies that can combat hazards like fire and gas leak with maximum efficiency. To optimize the effectiveness, the fundamental rule followed by all responders is the “wall-follower rule”. To avoid missing island rooms, two other methods can be chosen depending on hazard type. For hazard types that hamper communication, namely gas leakage, one to two extra responders will be sent towards the inner part of the building to detect island rooms; for other hazards that allows electronic communication, the commander system should be adopted to ensure minimum time spent while sweeping the whole building. A quantitative research can be conducted but this is often useless with limited information and time but is a great and methodical prophylactic way to plan in advance.

9. Letter to Committee

To: Local Emergency Planning Committee

From: HiMCM Team 17247

Date: November 19, 2025

Subject: Proposal for Optimized Evacuation Protocols in Complex Architectures

Dear Members of the Committee,

We hope this letter finds you well. We are writing to formally propose a modernized set of evacuation strategies tailored to the increasing complexity of today's urban infrastructure. As modern architecture trends toward taller but more intricate designs, the risks posed by hazards such as fire and gas leaks have evolved, threatening occupants who may be unable to escape in time. We believe that relying solely on traditional methods may no longer yield maximum efficiency. Based on our mathematical modeling of hazard dynamics, we propose the following protocols to enhance responder efficiency and occupant survival rates.

For standard building layouts, such as office buildings or schools featuring a central corridor with rooms on the outer edge, we recommend a streamlined "Counterclockwise Wall-Follower" strategy. We propose assigning two to three responders per floor who move continuously in a counterclockwise direction, checking each room sequentially. To ensure redundancy without sacrificing speed, it is vital that responders place a visual marker, such as a sticky note, on the door of each cleared room. This simple protocol allows subsequent responders to instantly recognize cleared areas, preventing unnecessary re-checks while maintaining a high safety standard.

However, our analysis confirms that this basic strategy is insufficient for complex, ring-shaped layouts common in shopping malls. In these scenarios, a standard wall-follower naturally neglects the inner edge of the corridor. Therefore, we recommend two resolution to cope with this situation. Firstly, increase the deployment to three or four responders per floor. To ensure total coverage, the team must be split: two to three responders should space themselves out along the outer wall moving counterclockwise, while one to two responders simultaneously trace the inner edge. This split-path approach ensures that central atriums and "island" rooms are cleared with the same rigor as storefronts. Secondly, if possible a commander should be deployed outside the building, and each in-building responder should carry a device that locates each of them and allows the commander to communicate with them (if the environment allows so), this way we can ensure that the responders have fully swept the room without ignoring entire "islands" and other parts of the building. In the back of the letter, we have made a flowchart for better visualizing and understanding the sweeping strategy, as well as for helping future improvements.

Furthermore, we strongly recommend a dynamic "Surge" resource allocation based on hazard proximity. Our findings suggest that the floor where a fire or gas leak originates, along with the floors immediately directly above and below, must be assigned at least twice the standard number of responders. Apart from the immediate hazard source, sweep teams should be trained to prioritize high-risk areas, specifically childcare rooms and laboratories, before clearing standard rooms. Finally, specific tactics must be employed for gas leaks and crowd management. When a gas leak is identified, two responders should start at the source room and sweep away from it, effectively herding occupants out of the danger zone rather than allowing them to backtrack into high concentrations. Additionally, to manage the chaos of moving crowds, one to two responders should remain in the hazard source zone specifically to guide evacuees, ensuring

the evacuation flow does not stagnate near the danger. We trust these actionable strategies will assist the committee in refining local emergency protocols.

Sincerely,

Team 17247 High School Mathematical Contest in Modeling

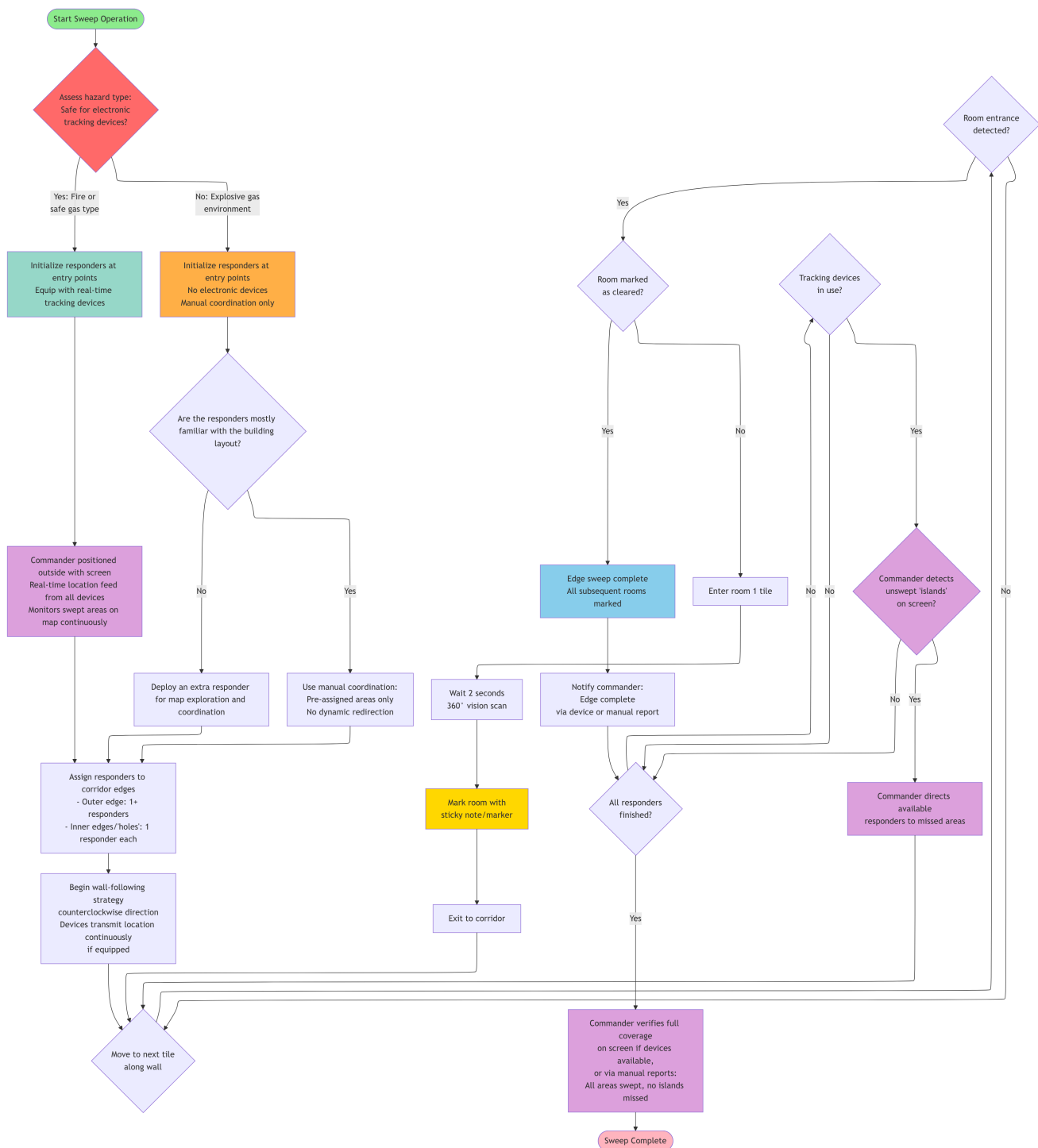


Figure 8: Full flowchart for responders after incorporating the refined model and the commander system

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Appendix

A Report on Use of AI Tools

- Perplexity AI was used to find some of the reference sources, its response was never directly copied in the paper
- Windsurf AI was used to assist writing the code for simulation and drawing the room layouts